

بسم الله الرحمن الرحيم, تحيه طيبه وبعد....

□ Description: This document contains token classes, a regular expression, and a DFA for a Tiny language compiler project.

>team number :30

>team members :

1- Youssef anwer abdel mohsen	Id: 2023170716
2- Youssef tamer moharam	Id:2023170719
3- Marwan Mohamed ahmed	Id:2023170572
4- Ahmed_Ali Mahdy	Id:2023170046
5- Yaseen Mohamed fathy	Id:2023170706
6- Ahmed Ayman Muhammed	Id:2023170025

>Token_Class

data type : enumerated type

```
{
    Int, Float, String, Read, Write, Repeat, Until, If, Elseif,
    Else, Then, Return, Endl, Main,

    Semicolon, Comma, LParanthesis, RParanthesis, LCurlyBracket, RCurlyBracket,

    AssignmentOp,
    EqualOp,
    LessThanOp,
    GreaterThanOp,
    NotEqualOp,
    ANDOp,
    OROp,

    PlusOp, MinusOp, MultiplyOp, DivideOp,

    Identifier,
    Constant,
    StringLit,
    Comment
}
```

> Letter := [a-zA-Z]

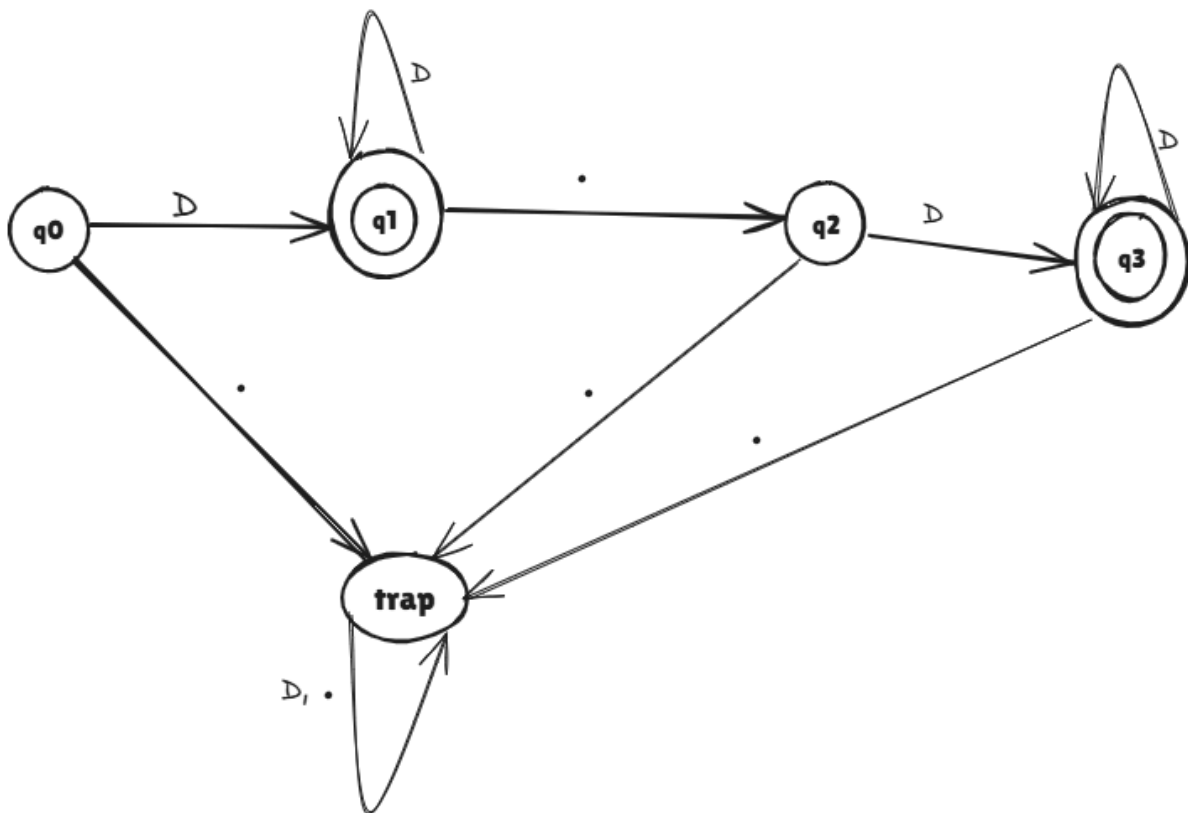
> Digit := [0-9]

1. Number:

> regex := {Digit}+(\.{Digit}+)?

> Alpha: {D , . }

>DFA:



2. String :

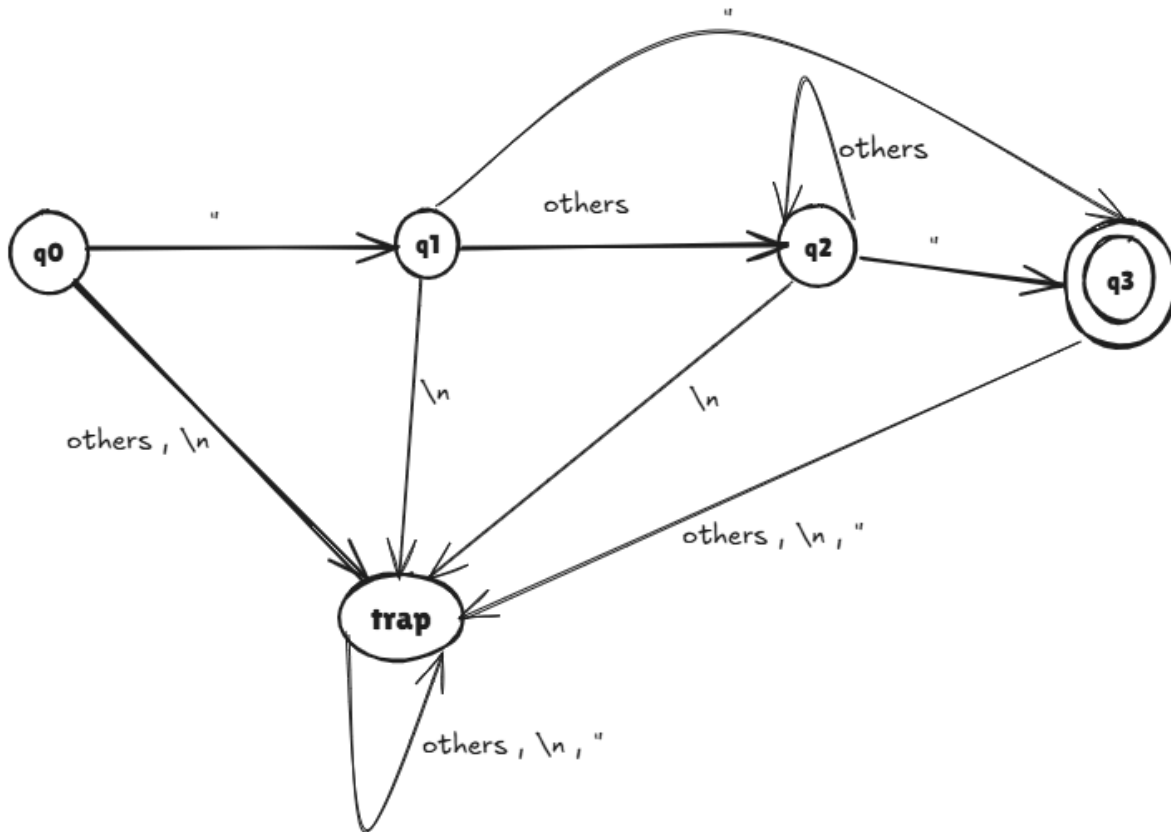
>regex := "([^\n])"

>Alpha: { " , \n , others }

> Others : any character except in Alpha.

> DFA :

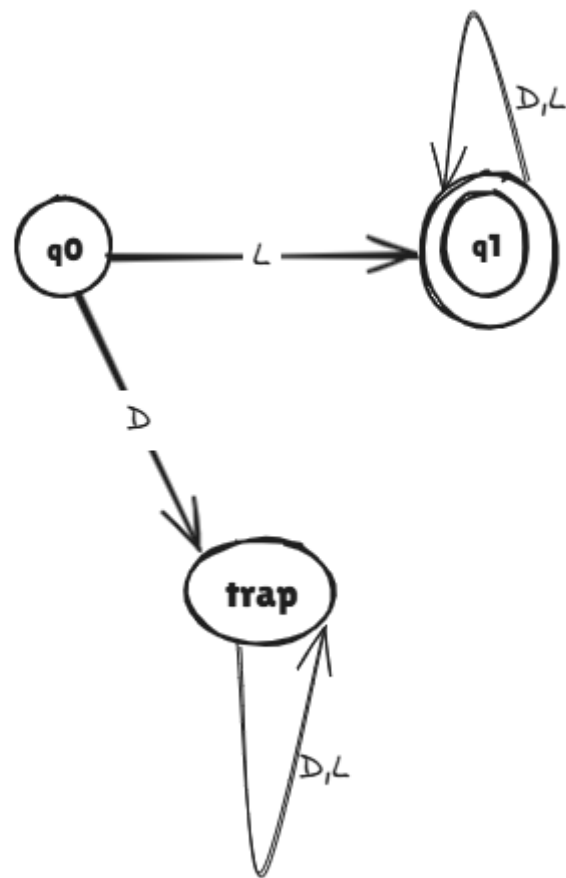
nvaf, hold [Scroll wheel] or [Space] while dragging, or use the hand tool



3. Identifier :

> regex: $\{Letter\}(\{Letter\}|\{Digit\})^*$
> Alpha: $\{L, D\}$

> DFA:



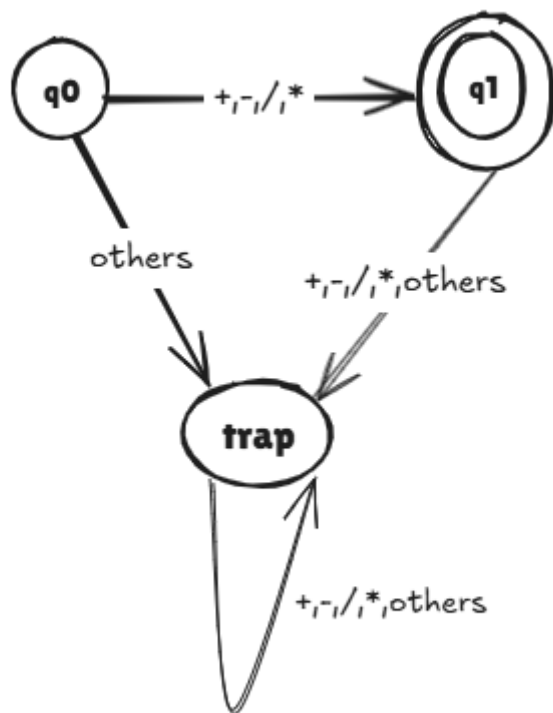
4. Arithmetic Operator:

> regex: $\backslash+|\backslash-|\backslash*|/$

> Alpha: $\{+, -, *, /, \text{others}\}$

> Others : any character except in Alpha.

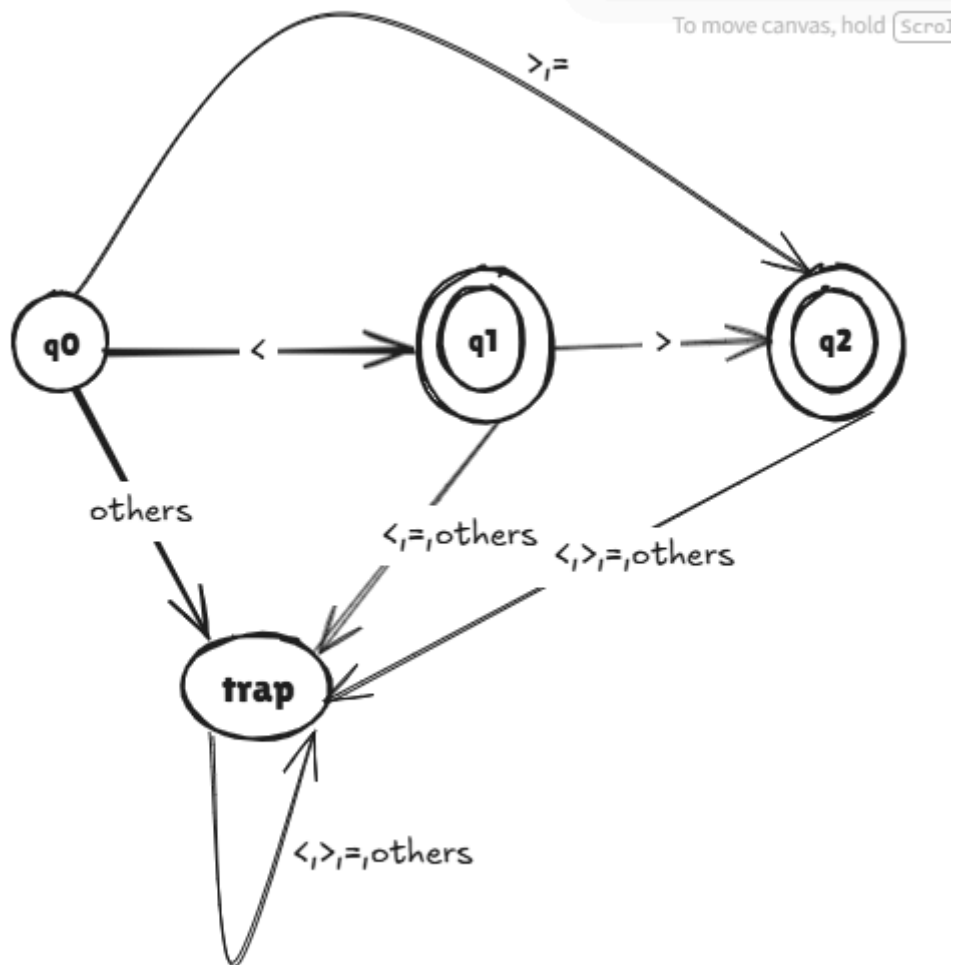
> DFA:



5. Condition Operator

- > regex := <|>|=|<>
- > Alpha: {<, >, =, <>, others}
- > Others : any character except in Alpha.

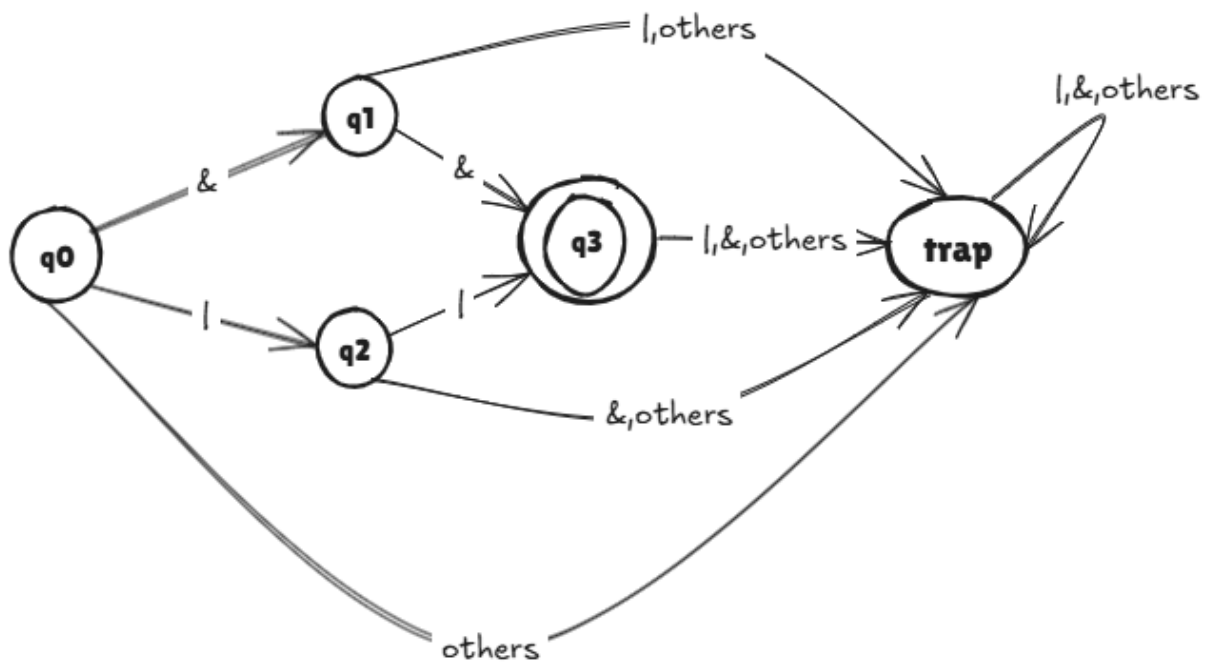
>DFA:



6. Boolean Operator:

- >regex := &&|\|
- >Alpha: {&, |, others}
- > Others : any character except in Alpha.

>DFA:



7. Comment Statement:

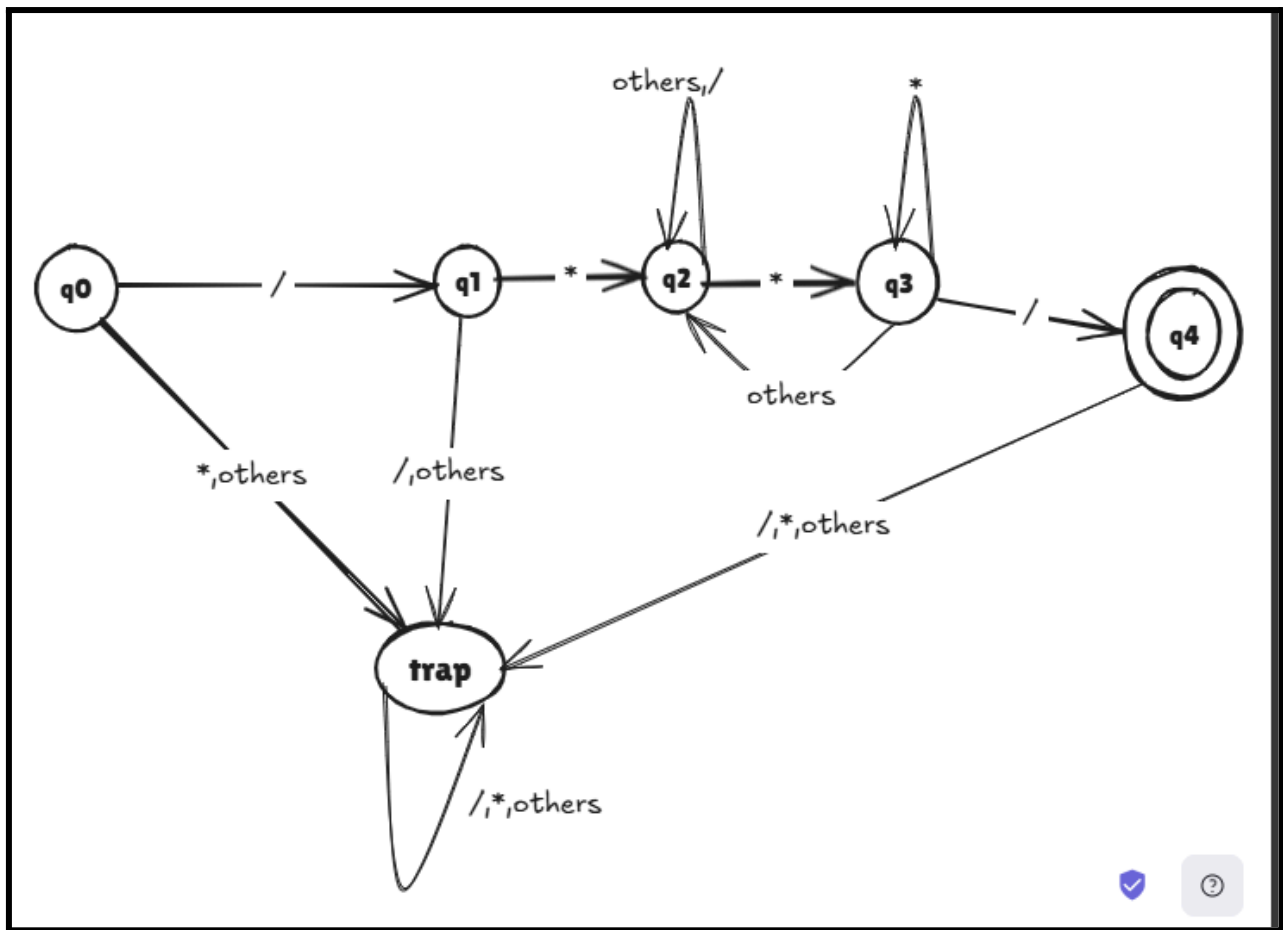
>regex :=

$\wedge^*([\wedge^*]\wedge^*[\wedge^*])\wedge^* /$

>Alpha: { *, /, others}

> Others : any character except in Alpha.

> DFA;



8 . Reserved Keywords:

>regex:= int|float|string|read|write|repeat|until|if|elseif|else|then|return|end|endl|main

> Alpha: [a-z]

>DFA:

